Supplementary Material: Beyond the Nearest Actor: A Scenario-Disjoint Audit of Current-Frame ODD-Context Proxies in WOMD

Anonymous WACV Datasets Track submission

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1. Extended Feature Confirmation Analysis

Table 1 presents the complete verification breakdown for all 13 development-frozen candidate features on the sealed internal holdout cohort ($N = 255,164$ frames, $2,804$ scenarios). Confidence intervals are pre-specified 95% paired scenario-block percentile bootstrap intervals ($B = 1,000$). For compactness, prefixes are omitted, and tp denotes third-party actors.

Table 1. Complete 13-Feature Holdout Confirmation Table.

Feature Name	Dev Effect (ρ)	Holdout Effect (ρ)	Holdout 95% CI	Status
tp_closing_pressure_sum	+0.1724	+0.1712	[+0.1532, +0.1903]	CONFIRMED
tp_closing_pressure_max	+0.0691	+0.0505	[+0.0325, +0.0692]	CONFIRMED
n_actors_50m	+0.0367	+0.0437	[+0.0193, +0.0679]	CONFIRMED
n_actors_70m	+0.0359	+0.0403	[+0.0157, +0.0642]	CONFIRMED
dist_nearest_road_edge_m	+0.0733	+0.0382	[+0.0148, +0.0617]	CONFIRMED
n_actors_30m	+0.0406	+0.0353	[+0.0123, +0.0601]	CONFIRMED
vulnerable_proportion_70m	+0.0265	+0.0155	[-0.0077, +0.0376]	DIRECTIONAL
tp_speed_std_mps	+0.0333	+0.0113	[-0.0114, +0.0320]	DIRECTIONAL
vehicle_proportion_70m	-0.0259	-0.0144	[-0.0366, +0.0090]	DIRECTIONAL
tp_nearest_dist_m	-0.0329	-0.0297	[-0.0456, -0.0121]	CONFIRMED
lane_heading_dispersion_50m	-0.0499	-0.0364	[-0.0604, -0.0118]	CONFIRMED
tp_mean_speed_mps	-0.0436	-0.0477	[-0.0677, -0.0269]	CONFIRMED
dist_nearest_crosswalk_m	-0.0782	-0.0845	[-0.1050, -0.0658]	CONFIRMED

2. Complete Scenario Profile Temporal Matrix

Table 2 details the scenario-level temporal associations across all 17 operational-context features on the sealed holdout ($N = 2,804$ scenarios) across target Peak, Area-Under-Curve (AUC), and Time-Exposed TTC (TET3). Cross-level classes are assigned from the sign and 95% CI support of the frame effect and the target-Peak effect only; AUC and TET3 are displayed as additional temporal dimensions and do not determine the class label.

Display labels map to machine-readable classifications as follows: Stable $\rightarrow$ CROSS_LEVEL_STABLE, Unsupported $\rightarrow$ UNSUPPORTED_BOTH, Frame-local $\rightarrow$ FRAME_LOCAL, Sign-discordant $\rightarrow$ DISCORDANT_SIGN, and Temporal-only $\rightarrow$ TEMPORAL_EXPOSURE_SPECIFIC.

3. Threshold Sensitivity and KPI Construct Validity

3.1. Threshold Sensitivity Analysis

Table 3 presents the post-lock sensitivity analysis over collision horizon thresholds $\tau \in \{2.0\text{s}, 3.0\text{s}, 5.0\text{s}\}$.

3.2. Development-Only KPI Construct Validity

Table 4 reports development-only convergent alignment between the continuous TTC-based severity score and within-dataset vehicle-response KPIs from KPI_CONSTRUCT_VALIDITY_V6.csv. The weak positive rank correlations and negative known-group effect sizes for the two DRAC summaries are directionally discordant. We therefore treat this axis as mixed supportive construct evidence, not external validation.

WACV #9876

Table 2. Complete 17-Feature Scenario Profile Temporal Matrix on Sealed Holdout.

Feature Name	Target Peak [95% CI]	Target AUC (ρ)	Target TET3 (ρ)	Class
dist_crosswalk	-0.1285 [-0.167, -0.091]	-0.1204	-0.1110	Stable
dist_stop_sign	-0.0265 [-0.062, +0.010]	+0.0058	-0.0535	Unsupported
dist_speed_bump	+0.0334 [-0.002, +0.067]	+0.0035	+0.0213	Unsupported
dist_road_edge	-0.0040 [-0.036, +0.031]	+0.0736	-0.0137	Frame-local
lane_heading_disp_50m	+0.0212 [-0.018, +0.057]	-0.0458	-0.0065	Sign-discordant
n_actors_30m	+0.0475 [+0.013, +0.082]	+0.0255	+0.0476	Stable
n_actors_50m	+0.0579 [+0.023, +0.092]	+0.0297	+0.0437	Stable
n_actors_70m	+0.0606 [+0.028, +0.096]	+0.0344	+0.0236	Stable
vehicle_prop_70m	-0.0005 [-0.039, +0.034]	+0.0231	-0.0353	Unsupported
vru_prop_70m	+0.0007 [-0.035, +0.039]	-0.0228	+0.0378	Unsupported
actor_entropy_70m	+0.0098 [-0.025, +0.048]	-0.0204	+0.0058	Unsupported
tp_nearest_dist	-0.0460 [-0.084, -0.008]	-0.0285	+0.0036	Stable
tp_mean_speed	-0.0591 [-0.097, -0.022]	-0.0322	-0.0038	Stable
tp_speed_std	-0.0021 [-0.038, +0.032]	+0.0262	+0.0254	Unsupported
tp_heading_disp	+0.0790 [+0.041, +0.116]	+0.0035	+0.0085	Temporal-only
tp_closing_pressure_max	+0.0528 [+0.014, +0.090]	+0.0753	+0.0362	Stable
tp_closing_pressure_sum	+0.1914 [+0.155, +0.230]	+0.2345	+0.0728	Stable

Table 3. Sensitivity to Collision Horizon Threshold τ (Sealed Holdout).

τ	Prevalence	M_P AP	$M_{P+E_{\text{all}}}$ AP	ΔAP	ΔAUROC
2.0s	1.50%	0.3161	0.2719	−0.0442	+0.0240
3.0s	5.61%	0.3224	0.3370	+0.0147	+0.0377
5.0s	17.97%	0.4972	0.5633	+0.0660	+0.0452

Table 4. Supportive Within-Dataset Vehicle-Response KPI Alignment on Development Split.

KPI Metric	Spearman ρ	Cohen's d
SDC hard deceleration p95	+0.1347	+0.3451
SDC min longitudinal acceleration	−0.1248	−0.1013
True-clearance DRAC p95	+0.0498	−0.0946
True-clearance DRAC maximum	+0.0524	−0.0372
SDC absolute jerk p95	+0.0306	+0.0395

4. Cohort Flow and Selection Provenance

4.1. Cohort Breakdown

The frozen preprocessing input contained 18,445 agent-state and 18,445 map-feature scenario partitions. The evaluation pipeline enumerated every discovered agent-state partition and applied no further scenario sampling before deterministic splitting. Dynamic-signal partitions were available for 12,783 scenarios, but signal variables were excluded from the primary 17-feature set. Upstream source-shard coverage was not retained as probability-sampling metadata; the evaluated cohort should therefore not be interpreted as a representative sample of the full public corpus.

The evaluated 18,445 scenarios (1,674,495 frames) were partitioned deterministically into:

- Training Partition: 12,828 scenarios (1,167,348 frames)
- Internal Validation Partition: 2,813 scenarios (255,983 frames)
- Development Cohort Total: 15,641 scenarios (1,423,331 frames)
- Sealed Internal Holdout Cohort: 2,804 scenarios (255,164 frames)

Split assignment utilized SHA256('womd_r2_split_v1|42|' + scenario_id)/2^64 with thresholds $[0.70, 0.85]$.

4.2. Dataset Attribution and License Notice

This research was conducted using the Waymo Open Motion Dataset (v1.3.1) provided by Waymo LLC. In strict compliance with the Waymo Dataset License Agreement, raw TFRecords and individual per-frame derived records are not distributed in this supplementary package. Users must obtain authorized access directly from the official Waymo portal (https://waymo.com/open).